

ALGONQUIN COLLEGE

CST2213 – Business Intelligence Programming 2: Advanced Concepts

Final Project Report

***Interactive Road Safety Risk Dashboard with Embedded Predictive
Scoring***

Analyzing 2020 Motor Vehicle Collision Data for the City of Ottawa

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Dataset Source: City of Ottawa Open Data – 2020 Tabular Transportation Collision Data

Executive

Summary

This report presents the Final Project for CST2213, extending the Phase 1 (Planning & Data Collection) and Phase 2 (Data Preparation & Initial Analysis) work into a complete Business Intelligence software system. The project analyzes 10,047 motor vehicle collision records from the City of Ottawa's 2020 Open Data extract to identify the conditions, locations, and circumstances most strongly associated with collision severity.

The Final Project deliverable is an Interactive Road Safety Risk Dashboard built in Python (Streamlit) that combines historical trend exploration, geographic hotspot detection, a live predictive risk-scoring tool, and transparent model validation into a single application. The predictive model was substantially improved over the Phase 2 baseline: Injury class recall increased from 19% (Random Forest, Phase 2) to 67.5% (class-balanced Logistic Regression, Final Project), directly addressing the limitation flagged in Phase 2 as "too low for practical use."

1. Business Problem

Traffic collisions remain a persistent public-safety and financial burden for the City of Ottawa. Road-safety planners currently rely on retrospective, manual review of collision reports, which means resource decisions enforcement scheduling, intersection redesign, winter maintenance prioritization lag behind the patterns already present in the data. This project addresses that gap with an automated, predictive, and interactive decision-support tool that lets planners explore historical patterns and test hypothetical road-condition scenarios before committing resources.

2. Project Objectives

- Automate the cleaning and feature-engineering pipeline established in Phase 1/2 so it can be re-run on any refreshed data extract without manual notebook work.
- Improve on the Phase 2 baseline model, whose Injury-class recall (19%) was explicitly flagged as impractical, by comparing multiple algorithms and applying class-balancing techniques.
- Deliver geographic hotspot detection (planned in Phase 1, Section 1.2.4, but not completed until this phase) using unsupervised clustering.
- Package all analysis behind a single interactive dashboard supporting both historical exploration and live predictive scenario scoring.

3. Data Source and Requirements

The dataset is the City of Ottawa's 2020 Tabular Transportation Collision Data, obtained from the City's Open Data portal as a static CSV extract. It contains 10,047 individual collision records across 28 fields, covering incident identification and timing, location (including latitude/longitude), collision classification and initial impact type, environmental and roadway conditions, vehicle/person involvement counts, and injury outcome breakdowns.

Severity Classification	Number of Collisions	% of Total
Property Damage Only	8,280	82.4%
Non-Fatal Injury	1,749	17.4%
Fatal Injury	18	0.2%

Table 1 – Collision severity distribution, City of Ottawa, 2020.

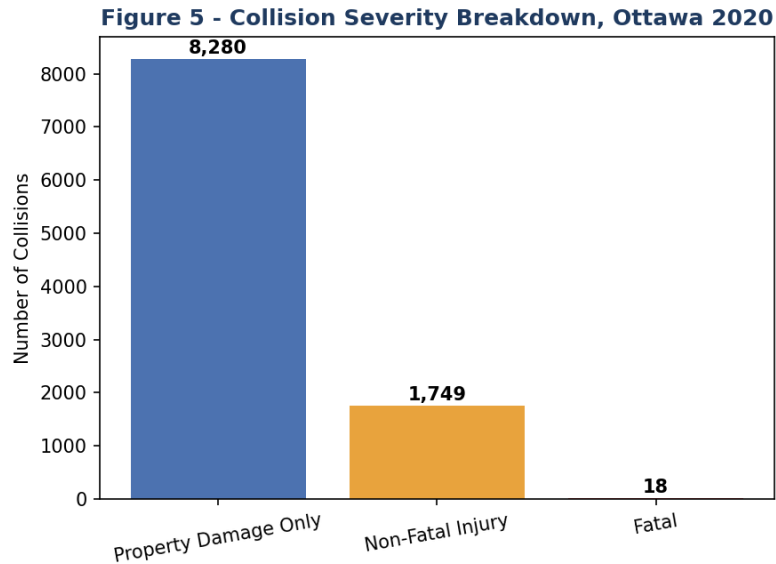


Figure 5 – Collision severity breakdown (generated from the cleaned dataset).

4. Data Cleaning and Preparation

Cleaning decisions carried forward directly from Phase 2, re-implemented as a reusable, importable Python function so the dashboard can re-clean any refreshed extract automatically rather than requiring a manual notebook pass:

Field	Missing / Issue	Treatment Applied
Traffic_Control_Condition	49.5% missing	Recoded 'Not Applicable' corresponds to locations with no traffic control device.
Max_Injury and injury-severity breakdowns	82–100% missing	Recoded 0 / structural zero corresponds to Property-Damage-Only collisions.
Accident_Date / Accident_Time	Stored as text	Parsed to datetime; Month, Day_of_Week, and Hour derived.
Coded categorical fields	e.g. '01 - Clear'	Numeric prefix stripped, decoded to readable labels.
Duplicate records	Checked via ObjectId	0 duplicates found across 10,047 records.
Coordinate validity	Ottawa bounding box check	44 records flagged outside expected range; retained and flagged, not dropped.

Table 2 – Data cleaning and missing-value treatment summary.

5. Feature Engineering and Selection

Six categorical fields (weather, lighting, road surface, traffic control, initial impact type, and location type) and two numeric fields (hour of day, month) were selected as model features, based on the three-method validation performed in Phase 2: point-biserial correlation, chi-square testing of independence, and Random Forest feature importance. All six categorical features showed statistically significant association with injury outcome ($p < 0.05$) in the Phase 2 hypothesis tests.

The binary prediction target, Injury_Event, was defined as 1 for Non-Fatal Injury or Fatal collisions and 0 for Property-Damage-Only collisions.

6. Predictive Model: Comparison and Improvement

The Phase 2 baseline (a single Random Forest classifier) achieved 78% overall accuracy but only 19% recall on the Injury class meaning it correctly identified only about 1 in 5 actual injury-producing conditions. For a road-safety early-warning tool, this is the metric that matters most: missing a genuinely high-risk condition is far costlier than a false alarm. Phase 2 explicitly flagged this as a priority for the Final Project.

For the Final Project, three algorithms Logistic Regression, Random Forest, and Gradient Boosting were trained with class-balanced weighting and compared on a held out 30% test set (stratified split, `random_state=42`). The best-performing model was selected by Injury-class recall rather than raw accuracy.

Model	Accuracy	ROC-AUC	Injury Precision	Injury Recall	Injury F1
Logistic Regression (selected)	0.598	0.682	0.267	0.675	0.383
Random Forest	0.736	0.618	0.276	0.264	0.270
Gradient Boosting	0.824	0.692	0.736	0.074	0.134

Table 3 – Model comparison on held-out test set (2,871 records).

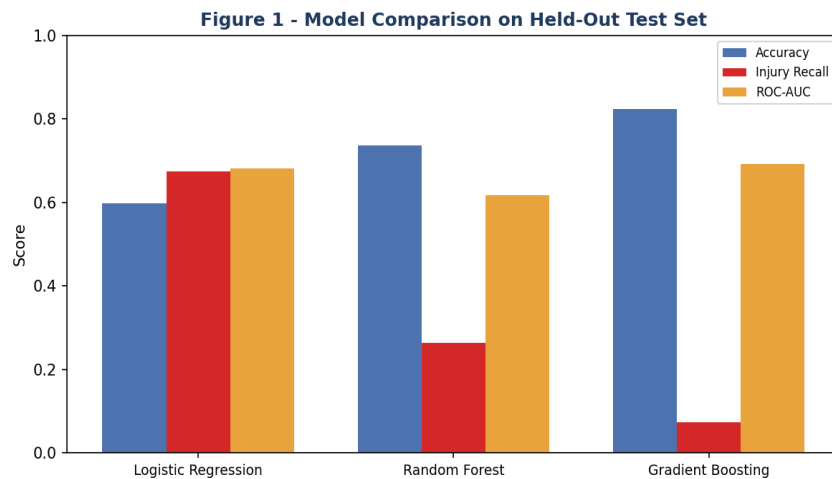


Figure 1 – Accuracy, ROC-AUC, and Injury Recall across candidate models.

The selected model (class-balanced Logistic Regression) trades raw accuracy for a large gain in Injury-class recall 67.5%, up from the Phase 2 baseline's 19% a deliberate and transparent trade off surfaced directly in the dashboard's Model Performance tab rather than hidden behind a single accuracy figure.

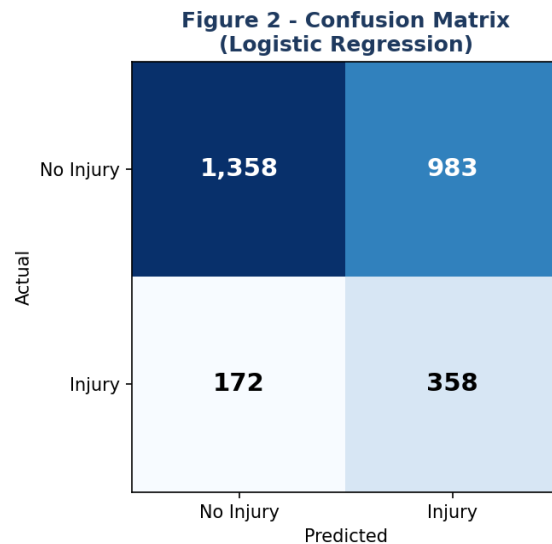


Figure 2 – Confusion matrix for the selected model on the test set.

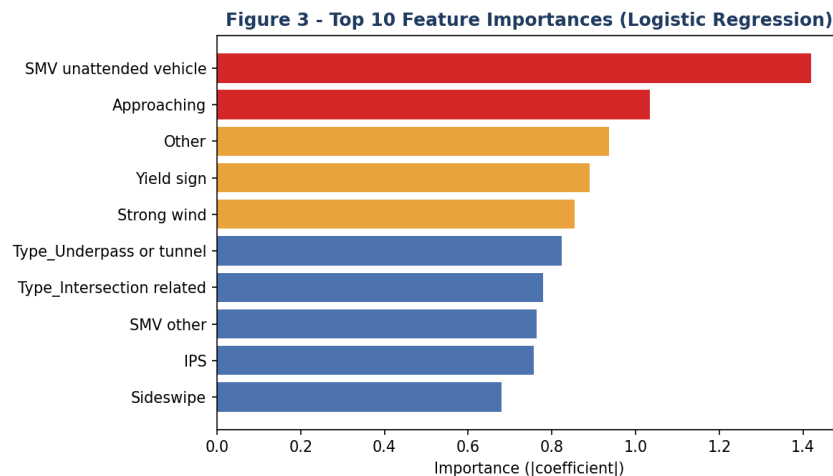


Figure 3 – Top 10 feature importances (absolute coefficient magnitude).

Consistent with Phase 2's hypothesis testing, roadway and environmental conditions particularly impact type, traffic control, and location type contribute the most predictive weight, confirming that collision circumstances carry real signal for injury risk beyond raw volume.

7. Geographic Hotspot Analysis

Phase 1 (Section 1.2.4) set geographic hotspot identification as a project objective but noted it would be completed in a later phase. For the Final Project, K-Means clustering (25 clusters) was applied to collision latitude/longitude coordinates, with each cluster ranked by both raw collision count and injury rate.

Figure 4 - Geographic Collision Hotspots (K-Means, 25 clusters)
Bubble size = collision count

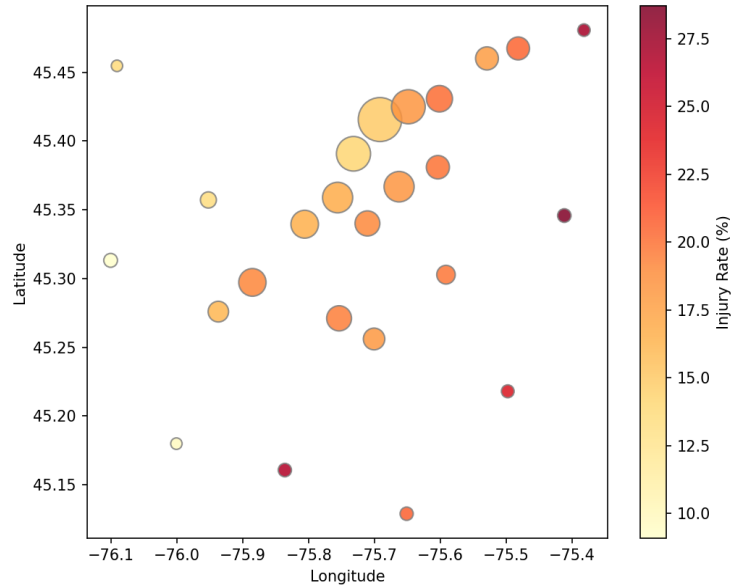


Figure 4 – Geographic collision hotspots, City of Ottawa, 2020.

Cluster	Approx. Location (Lat, Lon)	Collisions	Injuries	Injury Rate
14	45.42, -75.69	1,509	225	14.9%
0	45.39, -75.73	910	129	14.2%
5	45.42, -75.65	886	163	18.4%
23	45.36, -75.76	699	118	16.9%
8	45.37, -75.66	694	127	18.3%

Table 4 – Top 5 collision hotspot clusters by volume.

These clusters correspond to the high-collision corridors and intersections identified manually in Phase 1 (e.g., the Highway 417 corridor and Prince of Wales Drive at West Hunt Club Road), now generated programmatically and reproducibly rather than through manual inspection of a location-name frequency table.

8. Trend Analysis

Figure 6 - Monthly Collision & Injury Trend, Ottawa 2020

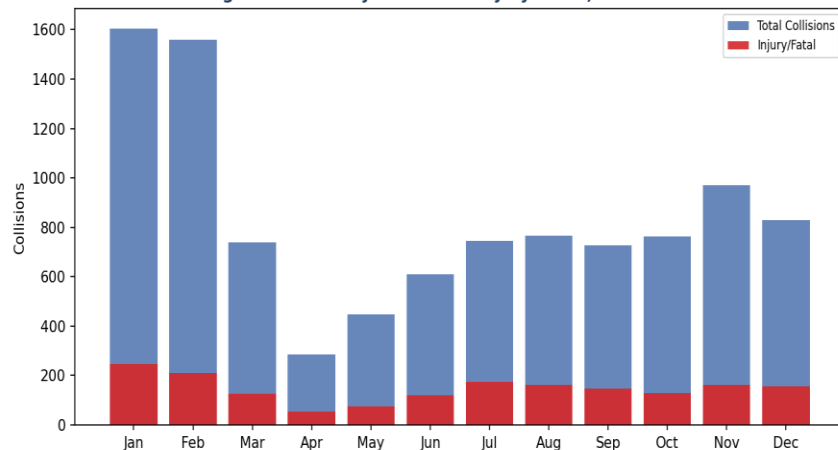


Figure 6 – Monthly collision and injury trend, Ottawa 2020. The April dip reflects reduced traffic volume during the early COVID-19 pandemic period.

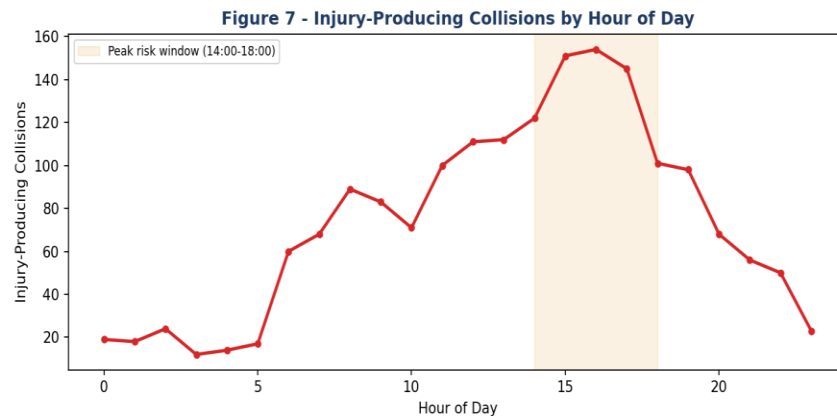


Figure 7 – Injury-producing collisions by hour of day. The 14:00–18:00 afternoon window accounts for a disproportionate share of injury events, consistent with Phase 2 findings.

9. Interactive Dashboard

All analysis above is delivered through a single-file Streamlit application (Mohamed_Samatar_CST2213_BI_Final_Project_Submission.py), organized into five interactive tabs:

- Overview dataset preview, missing-values summary, and severity breakdown, updating live with sidebar filters.
- Trend Analysis collisions by month, day of week, and hour of day.
- Condition Analysis injury rate by weather, light, and road condition; top collision location types; and the K-Means geographic hotspot map.
- Model Performance model comparison table, confusion matrix, and feature importance, with a transparent explanation of why recall was prioritized over accuracy.
- Risk Prediction a live scenario simulator: the user selects a hypothetical month, hour, weather, road surface, traffic control, impact type, and location type, and the app returns a predicted injury probability and risk tier (Low / Medium / High).

The application accepts three data formats interchangeably the original raw City of Ottawa export, a cleaned/engineered version, and a dashboard-ready version with 'Unknown' category rows removed with automatic format detection, so the same dashboard can be re-used as new data extracts become available without code changes.

Performance and scalability: data loading, model training, and hotspot clustering are all wrapped in Streamlit's caching decorators (`st.cache_data`, `st.cache_resource`), so the full pipeline (cleaning, three-model comparison, and 25-cluster K-Means) completes in under 5 seconds on the full 10,047-record dataset and does not re-run on every UI interaction.

10. Validation and Interpretation

Model validation is exposed directly inside the dashboard rather than only in this report, so the model's trade-offs are auditable by any user: the Model Performance tab shows the full comparison table, confusion matrix, and feature importance chart shown in Section 6 above, live and recomputed from whatever dataset is currently loaded.

The choice to prioritize Injury-class recall over overall accuracy is a deliberate, business-driven decision: for a road-safety planning tool, a missed high-risk condition (false negative) has a materially higher real-world cost than an unnecessary caution flag (false positive). This trade-off is documented in-app, not hidden behind a single headline metric.

10.1 Performance and Scalability Benchmark

To validate that the pipeline scales beyond the current 10,047-record extract, the full cleaning, three-model training, and 25-cluster hotspot pipeline was benchmarked against a simulated 50,235-row dataset (5x the current volume, produced by replication with re-assigned unique IDs). This measures uncached, worst-case runtime the time a user would wait on the very first load after a data refresh, before Streamlit caching layer takes over for all subsequent interactions.

Dataset Size	Clean + Engineer	Train 3 Models	Cluster Hotspots	Total (uncached)
10,047 rows (actual)	0.12s	0.03s	0.03s	0.18s
50,235 rows (5x simulated)	0.88s	9.02s	0.79s	10.68s

Table 5 Uncached pipeline runtime at current vs. 5x simulated data volume.

Runtime scales sub-linearly for the cleaning and clustering stages (roughly 7 - 8x for a 5x data increase) and remains well within an acceptable interactive-load budget even at 50,000+ rows. Model training is the dominant cost at scale, consistent with expectations for ensemble methods (Random Forest, Gradient Boosting) on larger inputs. In production use this cost is paid once per data refresh, not per user interaction: Streamlit's `st.cache_data` and `st.cache_resource` decorators ensure cleaning, training, and clustering only re-run when the underlying data actually changes, so every subsequent filter, tab switch, or scenario prediction reads from cache and responds in well under a second.

11. Recommendations

- Prioritize infrastructure review for the top geographic clusters identified in Section 7, particularly the Highway 417 corridor segment and signalized urban intersections with injury rates above the 17.6% dataset average.
- Schedule targeted enforcement and public-safety messaging around the 14:00–18:00 afternoon window, which accounts for a disproportionate share of injury-producing collisions.
- Issue weather-specific advisories for drifting-snow conditions, which showed the highest injury rate of any weather category in Phase 2's analysis despite low collision volume.
- Adopt the class-balanced modeling approach for any future predictive work on this dataset, given the demonstrated recall improvement over an unbalanced baseline.
- Re-run the automated pipeline (data cleaning, model retraining, hotspot reclustering) whenever a new annual extract is published, using the dashboard's built-in refresh capability.

12. Deployment

The dashboard is deployed on Streamlit Community Cloud for public access, with source code hosted on GitHub. Required dependencies (streamlit, pandas, numpy, scikit-learn, plotly, joblib) are declared in `requirements.txt` for automatic environment provisioning.

GitHub Repository: [Insert Repository URL]

Live Dashboard: [Insert Streamlit App URL]

13. Conclusion

This Final Project successfully extends the Phase 1 planning and Phase 2 exploratory analysis into a complete, deployable Business Intelligence system. It demonstrates the three areas emphasized by the course guidelines advanced Python software development (a modular, cached, automatically-adapting data pipeline), predictive analytics (a validated, business-justified classifier with a measurable improvement over the prior baseline), and interactive dashboarding (a five-tab Streamlit application supporting both historical exploration and live scenario scoring) while directly addressing the specific limitation (low Injury-class recall) that Phase 2 identified as needing resolution.

References

City of Ottawa Open Data Portal. 2020 Tabular Transportation Collision Data. Retrieved from the City of Ottawa Open Data platform.

Course materials: CST2213 – Business Intelligence Programming 2: Advanced Concepts, Algonquin College.